Building for one, building for all

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From Playlists to Policy: Automating the Analysis of Thousands of Hours of Public Meetings

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Problem Statement

- People spend hours transcribing public meetings
- Videos may require accurate closed captions for accessibility (WCAG 2.1 Level AA or Section 508)
- Videos may not be accessible for non-native speakers
- Human transcription may not be feasible for large video archives
- Valuable civic data remains "locked" inside long-form video
- Disconnected services for transcription and language access

Transcript

0:00 you're live Okay Thank you and welcome to the AIB blueprint for Maryland's

0:06 AIB Today's date is uh May the 15th and

0:11 we're meeting officially about 10:05 p.m call the meeting to the order Call

0:17 the meeting to order And first item for us is a couple of open comments Let me thank our staff uh for all the

0:24 tremendous work uh throughout the entire legislative session uh it was rather busy Uh we had some

0:32 real challenges and I think that uh given the circumstances and conditions

0:37 uh we probably weathered those probably as well as we could Uh there are still some additional challenges that are out

0:44 there and we will have an opportunity to review those as we go forward But uh given where we are today I think that

0:52 the AIB came out of it somewhat okay in my opinion Um first of all let's adopt

0:58 the last meeting minutes summary Uh hope that we had an opportunity to review

English (auto-generated)

AIB Regular Meeting - May 15, 2025

Blueprint Accountability and Implementation Board
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Live chat replay was turned off for this video.

Solutioning through intake I

- IT procurements and software review at the State of Maryland are legislatively required to go through the Department of IT (DoIT) (authority with Office of State Procurement).

- DoIT acts as a service provider to understand problem statements, avoid IT sprawl and ensure we build/buy secure services and applications that are cost effective

Solutioning through intake II

"I would like to purchase XYZ to help transcribe public meetings"

"XYZ is not approved due to privacy concerns of it running in our tenant. Can we meet to learn more about your use case?"

Building a transcription analysis service

- Large open-source developer ecosystem around YouTube
- Easy to build upon

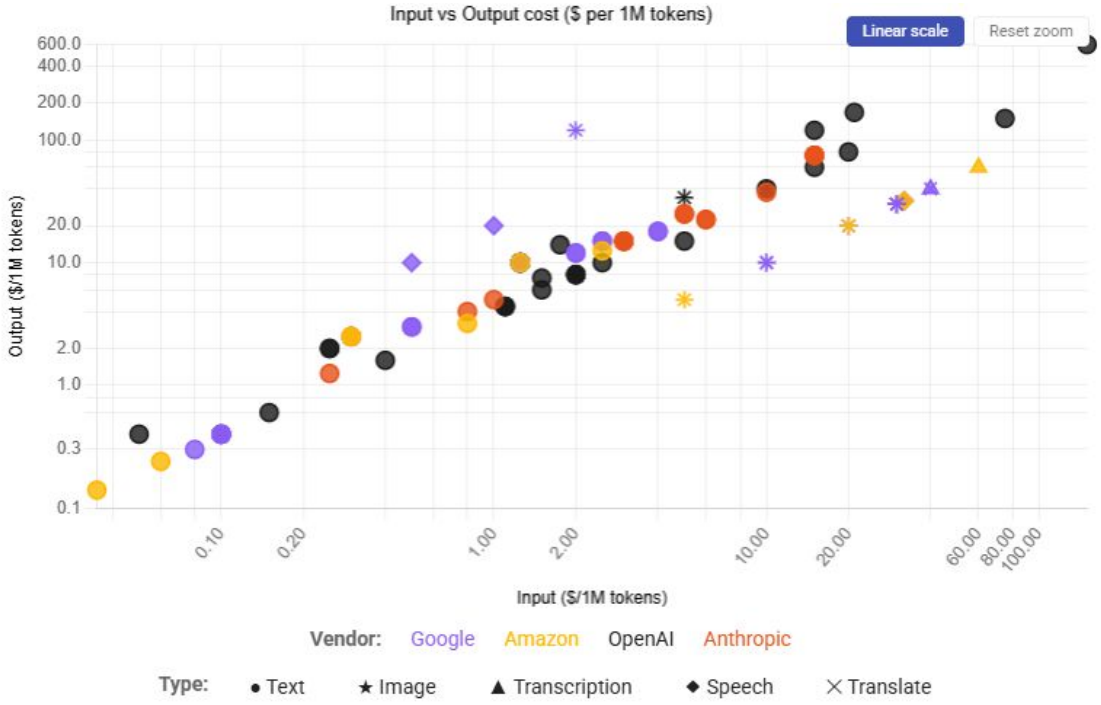
>_ dlp

YT-DLP *A feature-rich command-line audio/video downloader*

```
  {'duration': 4.96,
   'start': 11476.479,
   'text': 'with you all again soon. Have a great'},
  {'duration': 5.04, 'start': 11478.16, 'text': 'afternoon, everyone.'},
  {'duration': 3.92, 'start': 11481.439, 'text': '>> Thanks.'},
  {'duration': 4.56,
   'start': 11483.2,
   'text': '>> Thank you, staff. Appreciate you all so'},
  {'duration': 2.401, 'start': 11485.359, 'text': 'much.'}]]
PS C:\Users\ray.bell> uvx --from youtube-transcript-api youtube_transcript_api atmGAHYpf_c
```

Vendor agnostic

- DoIT manages:
  - Google Workspace
  - Microsoft 365
  - AWS
  - GCP
  - Azure



- Tool designed to meet agencies where they are at
- Options to choose between various models/products (Cloud vs. LLM)

YouTube To Docs: Developer tool

- Framework for users to run on their local machine
- Not designed to be deployed:
  - Handling of client id and secrets to write to workspace and sharepoint
  - Cloud IP’s are often blocked from crawling YouTube
  - Limit exposure of API keys

Version 1

DoIT-Artificial-Intelligence / youtube-to-docs

YouTube → Google colab → Docs → Gemini

```
pip install youtube-transcript-api
```

Summarize with custom instructions, such as spelling acronyms correctly

Version 2

DoIT-Artificial-Intelligence / youtube-to-docs

```
PS C:\Users\ray.bell> uvx youtube-to-docs --help
Installed 57 packages in 1.72s
Usage: python.exe C:\Users\ray.bell\[redacted]\Scripts\youtube-to-docs [-h] [-o OUTFILE] [-t TRANSCRIPT]
                                                                        [-m [MODEL]] [--tts TTS]
                                                                        [-i [INFOGRAPHIC]]
                                                                        [--alt-text-model ALT_TEXT_MODEL]
                                                                        [-nys] [-tr TRANSLATE] [-cia] [-a ALL]
                                                                        [--verbose]
                                                                        [-scc [SUGGEST_CORRECTED_CAPTIONS]]
                                                                        [-pp POST_PROCESS]
                                                                        [video_id]

Positional Arguments:
  video_id              Can be one of: A YouTube URL e.g. `https://www.youtube.com/watch?v=atmGAHYpf_c` A Video ID e.g. `atmGAHYpf_c` Playlist ID (starts
                        with `PL` e.g. `PL8ZxoInteClyHaiReuOHpv6Z4SPrXtYtW`) Channel Handle (starts with `@` e.g. `@mga-hgo1740`) Comma-separated list of
                        Video IDs. (e.g. `KuPc06JgI_A,GalhDyf3F8g`)

Options:
  -h, --help            show this help message and exit
  -o, --outfile OUTFILE
                        Can be one of: Local file path to save the output CSV file. `workspace` or `w` to store to Google Drive or a workspace folder ID.
                        `sharepoint` or `s` to store to Microsoft SharePoint. `memory` or `m` to keep artifacts in memory (no files on disk). `none` or
                        `n` to skip saving to a file (results will be in the log).
  -t, --transcript TRANSCRIPT
                        The transcript source to use. Can be `youtube` (default) to fetch existing YouTube transcripts, AI model name (e.g.
                        `gemini-3.1-pro-preview`, `gemini-3-flash-preview`, `gcp-chirp3`, perform STT on extracted audio. For `gcp-` models (Cloud
                        Speech-to-Text V2), `GOOGLE_CLOUD_PROJECT` is required and `YTD_GCS_BUCKET_NAME` is optional/recommended. For `aws-transcribe`,
                        `YTD_S3_BUCKET_NAME` is required.
  -m, --model [MODEL]   The LLM to use for speaker extraction, Q&A generation, tag generation, and summarization. Can be one of: Gemini model (e.g.,
                        `gemini-3-flash-preview`) GCP Vertex model (prefixed with `vertex-`; e.g. `vertex-claude-haiku-4-5@20251001`) AWS Bedrock model
                        (prefixed with `bedrock-`; e.g. `bedrock-claude-haiku-4-5-20251001-v1` `bedrock-nova-2-lite-v1`) Azure Foundry model (prefix with
                        `foundry-`; e.g. `foundry-gpt-5-mini`) Can also be a comma-separated list of models (e.g.
                        `gemini-3-flash-preview,bedrock-claude-haiku-4-5-20251001-v1`). Defaults to `None` (Transcript only).
```

Version 2.1

DoIT-Artificial-Intelligence / youtube-to-docs

MCP and SKILLS

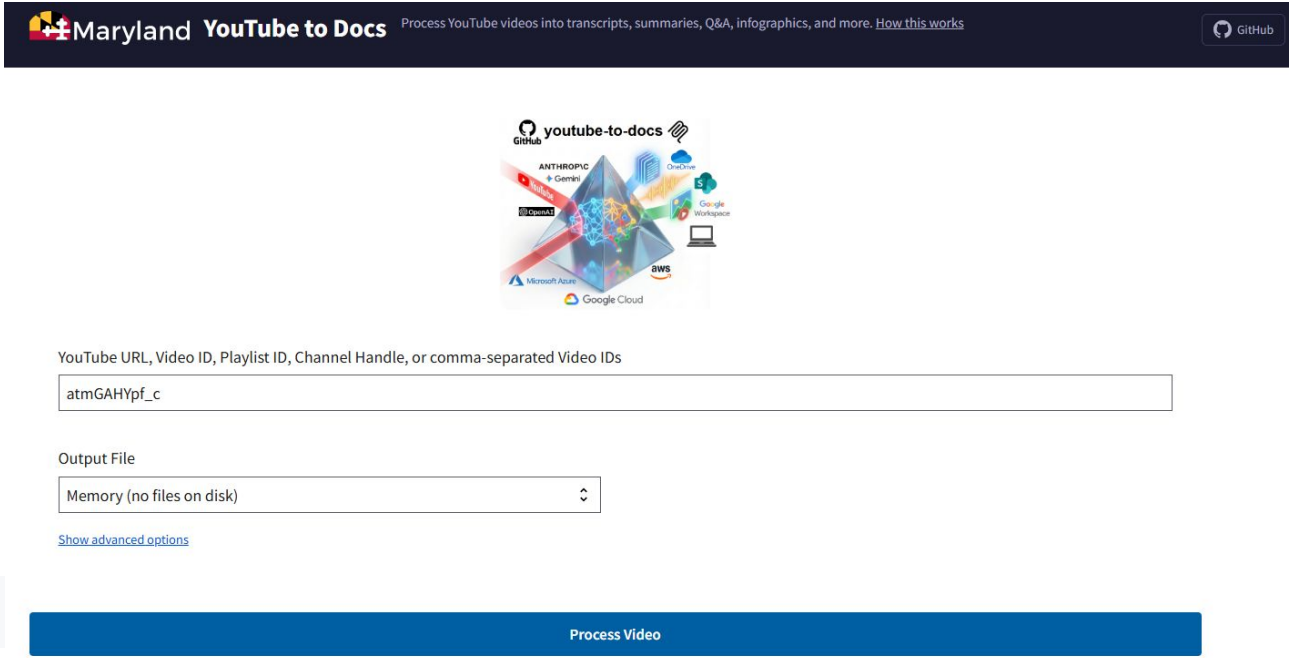
gemini extensions install
https://github.com/DoIT-Artificial-Intelligence/youtube-to-docs.git

/plugin marketplace add DoIT-Artificial-Intelligence/youtube-to-docs
/plugin install youtube-to-docs@youtube-to-docs

Version 3

```
PS C:\Users\ray.bell> uvx --from "youtube-to-docs[all]" youtube-to-docs-app
Installed 113 packages in 3.82s
App available at: http://localhost:8000
INFO:     Started server process [27860]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

Maryland Web Design System



DoIT-Artificial-Intelligence / youtube-to-docs

Agentic Engineering

- Employ best practices for Software engineering from the start:
  - Testing (Start with the tests)
  - Type checking (ty)
  - Linting and formatting (ruff)
  - Accessibility (axe and html validate)
  - Scripts to hand-check (bash and python files)
- Biggest wins
  - Using MDWDS library, Accessibility testing and API for cloud services
- Biggest losses
  - Repo drift
- Use with caution. Use approved tools. Watch costs.

Demo(s)

Example speaker identification to improve Closed Caption text

youtube generated - B0x2I_doX9o - PHS Committee Session, 4_

youtube-to-docs-test > srt-files > youtube generated - B0x2I_doX9o - PHS Co

```
1
00:00:02,639 --> 00:00:09,040
Good morning. Uh thank you. It is uh

2
00:00:05,680 --> 00:00:10,639
Monday, April 7th. Happy siny die. This

3
00:00:09,040 --> 00:00:13,518
is public health and minority health

4
00:00:10,639 --> 00:00:15,759
disparity subcommittee. We have one bill
```

gemini-3-flash-preview suggested corrections - B0x2I_doX9o - PHS Committee Session, 4_7_2025 #1.srt

youtube-to-docs-test > suggested-corrected-caption-files > gemini-3-flash-preview suggested correctio

```
1
00:00:02,639 --> 00:00:15,759
[Chair Bagnel] Good morning. Uh, thank you. It is, uh, Monday, April 7th.

5
00:00:13,519 --> 00:00:48,719
We have one bill, um, for discussion today. It is Senate Bill 447. This is

15
00:00:45,280 --> 00:00:54,000
[Delegate Alustin] Oh, sorry. All of my questions have been answered. Than

19
00:00:52,480 --> 00:01:18,799
[Chair Bagnel] All right. Thank you very much, Delegate Alustin. Um, uh, s
```

Demo(s)

Question & Answer extraction from a YouTube video

question number	questioner(s)	question	responder(s)	answer	timestamp	timestamp url
1	UNKNOWN (Committee Member)	Regarding the statement about working with individuals "willing to work with us," who specifically within the sphere is not willing to work with the Office of Security Management?	James Saunders (Acting Chief Information Security Officer); Katie Savage (DoIT Secretary)	There is a mix of agencies; some maintain degrees of independence due to partially successful centralization efforts 10 years ago. The goal is to work with the willing first and leverage law/legislation to bring others on board for centralized visibility. Secretary Savage added that larger agencies tend to be less "enterprised," creating technology sprawl that requires relationship building and addressing legacy infrastructure.	00:27:02	https://www.youtube.com/watch?v=WTC39xMj4-k&t=1622
2	UNKNOWN (Committee Member)	Have any improvements been made to make the CISO the ultimate decision-making agency in the state during a cyber attack, effectively overriding pushback from agencies claiming federal system independence?	James Saunders (Acting Chief Information Security Officer)	Yes, the State CISO has that power. It is clear in law and was executed clearly multiple times over the past summer.	00:29:10	https://www.youtube.com/watch?v=WTC39xMj4-k&t=1750
3	UNKNOWN (Committee Member)	Do you work with counties and local municipalities, who often struggle with costs and legacy systems (like Windows 10), to bring them in line or provide guidance?	James Saunders (Acting Chief Information Security Officer)	Yes, DoIT is working with 23 counties and Baltimore City, including developing a new policy suite with their input and engaging through the Maryland Local Cybersecurity Collaborative. They are working to extend state resources and capabilities to locals, while respecting Home Rule.	00:30:26	https://www.youtube.com/watch?v=WTC39xMj4-k&t=1826

Demo(s)

Translation from English to Spanish for a YouTube video

youtube-to-docs-artifacts > srt-files > youtube generated - fd1rMbv5loE - EBA

```
1
00:00:00,000 --> 00:00:03,000
Either

2
00:00:06,879 --> 00:00:09,879
way,

3
00:04:02,000 --> 00:04:06,239
Good afternoon everyone. If you could u

4
00:04:04,480 --> 00:04:08,079
take a seat please so we can get started

5
00:04:06,239 --> 00:04:10,319
```

youtube-to-docs-artifacts > srt-files > youtube translated gcp-translate-es - fd1

```
1
00:00:00,000 --> 00:00:03,000
De cualquier manera

2
00:00:06,879 --> 00:00:09,879

3
00:04:02,000 --> 00:04:06,239
Buenas tardes a todos. Si pudieran

4
00:04:04,480 --> 00:04:08,079
tomar asiento, por favor, para que podamos empezar.

5
00:04:06,239 --> 00:04:10,319
con nuestra audición, se lo agradecería mucho.
```

Demo(s)

Example Infographic of a YouTube video

Demo(s)

Metadata and summaries of a playlist

Description	Data Published	Channe	Tags	Duratio	Transcr	Transcr	One Se	One Sentence Summary bedrock-nova-2-lite-v1 from youtube
Health and Human Services Subcommittee Other for Tuesday, March 10, 2026 @ 1:30 PM	2026-03-11T06:23:30Z	MGA - B&T		0:59:52	https://sta	27,393	https://sta	The HHS Committee reviewed and decided on budgetary and policy recommendations across multiple state agencies, adopting various modifications, restrictions, and requests for reports while mostly concurring with the governor's allowances and agency agreements.
Health and Human Services Subcommittee Budget Hearing for Friday, March 6, 2026 @ 11:30 AM	2026-03-07T09:27:16Z	MGA - B&T		2:30:03	https://sta	116,959	https://sta	The Health and Human Services Committee Session on March 6, 2026, discussed budget changes and key issues for four agencies—MHBE, HRC, DHS CSA, and OHIP—including increased budgets driven by federal policy changes, transitions to new healthcare models, performance metrics, and energy assistance trends, while also addressing public concerns regarding trauma care funding, school mental health services, and child support services staffing and retention.
Health and Human Services Subcommittee Budget Hearing for Thursday, March 5, 2026 @ 1:00 PM	2026-03-06T08:09:16Z	MGA - B&T		1:28:56	https://sta	81,154	https://sta	The Maryland Department of Health's Behavioral Health Administration budget hearing for fiscal year 2027 highlighted a 3.8% decrease to $3.9 billion, with key concerns including increasing utilization of services, a projected Medicaid deficit, workforce shortages, the need for a 3% Medicaid reimbursement rate increase, implementation of the Assisted Outpatient Treatment program, and securing funding for Certified Community Behavioral Health Clinics, all underscoring the critical need for sustained investment in behavioral health services in Maryland.
Health and Human Services Subcommittee Budget Hearing for Monday, March 2, 2026 @ 1:00 PM	2026-03-03T07:38:28Z	MGA - B&T		1:34:04	https://sta	73,607	https://sta	The Health and Human Services Subcommittee meeting on March 2, 2026, focused on the Maryland Department of Health's $14 billion Medicaid budget for fiscal year 2027, highlighting performance improvements, ongoing challenges such as application delays and federal legislative changes (especially HR1), and calls from public witnesses for increased funding, better IT systems, and additional staff to protect coverage and access to care for vulnerable populations.

Try it yourself!

https://github.com/DoIT-Artificial-Intelligence/youtube-to-docs

uvx youtube-to-docs --help

uvx --from "youtube-to-docs[all]" youtube-to-docs-app